CS3011-INTRODUCTION TO ROBOTICS

Lecture 14







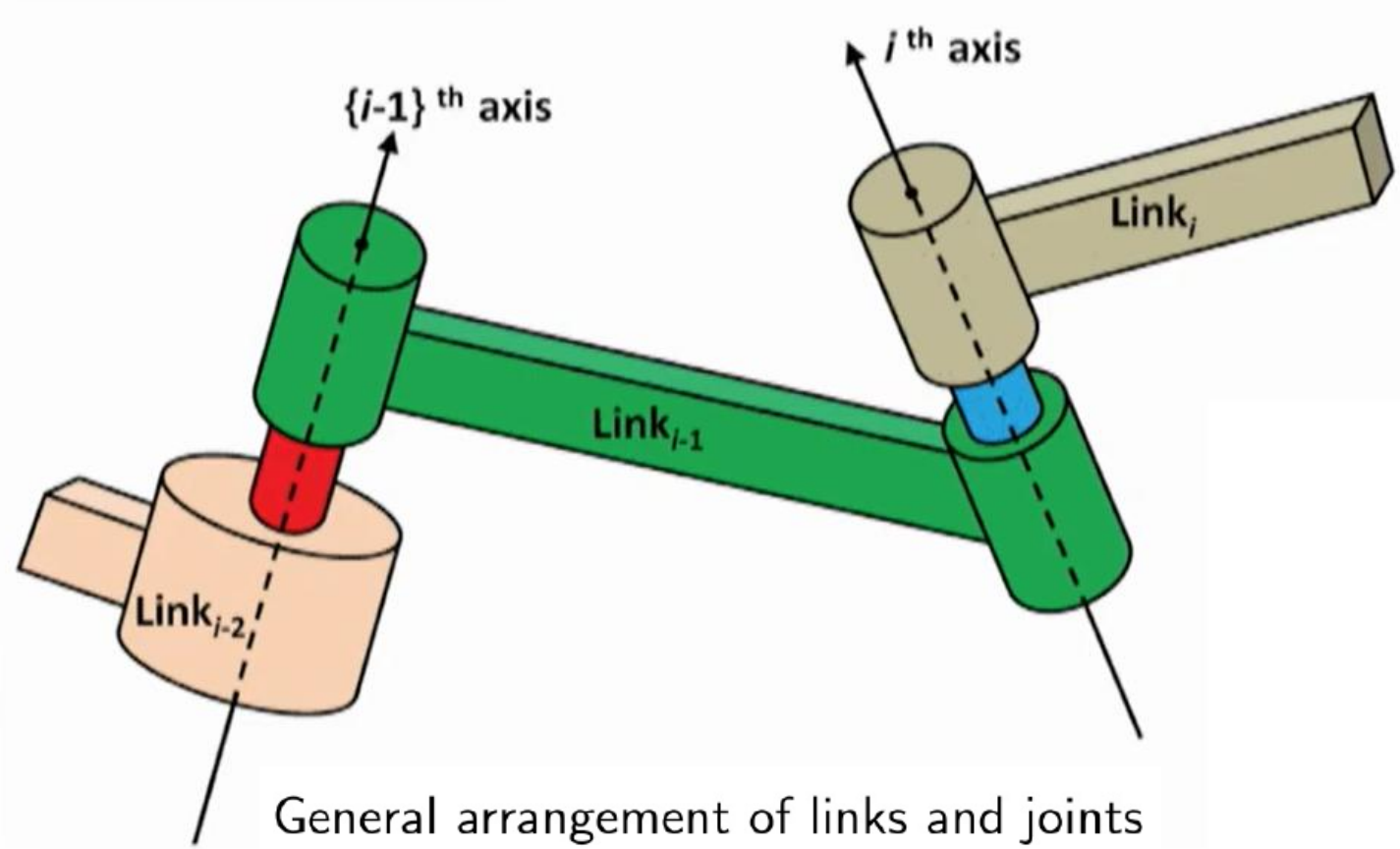
DH approach

- **Jacques Denavit and Richard Hartenberg** introduced a convention in 1955 in order to standardize the coordinate frames for spatial linkages.
- The kinematic parameters (also called DH parameters) are the **four parameters** associated with a particular convention for attaching reference frames to the links of a spatial kinematic chain, or robotic manipulator.
- In this convention, coordinate frames are attached to the joints between two links such that **one set of transformation is associated with the joint parameters**, and the **second set is associated with the link parameters**.

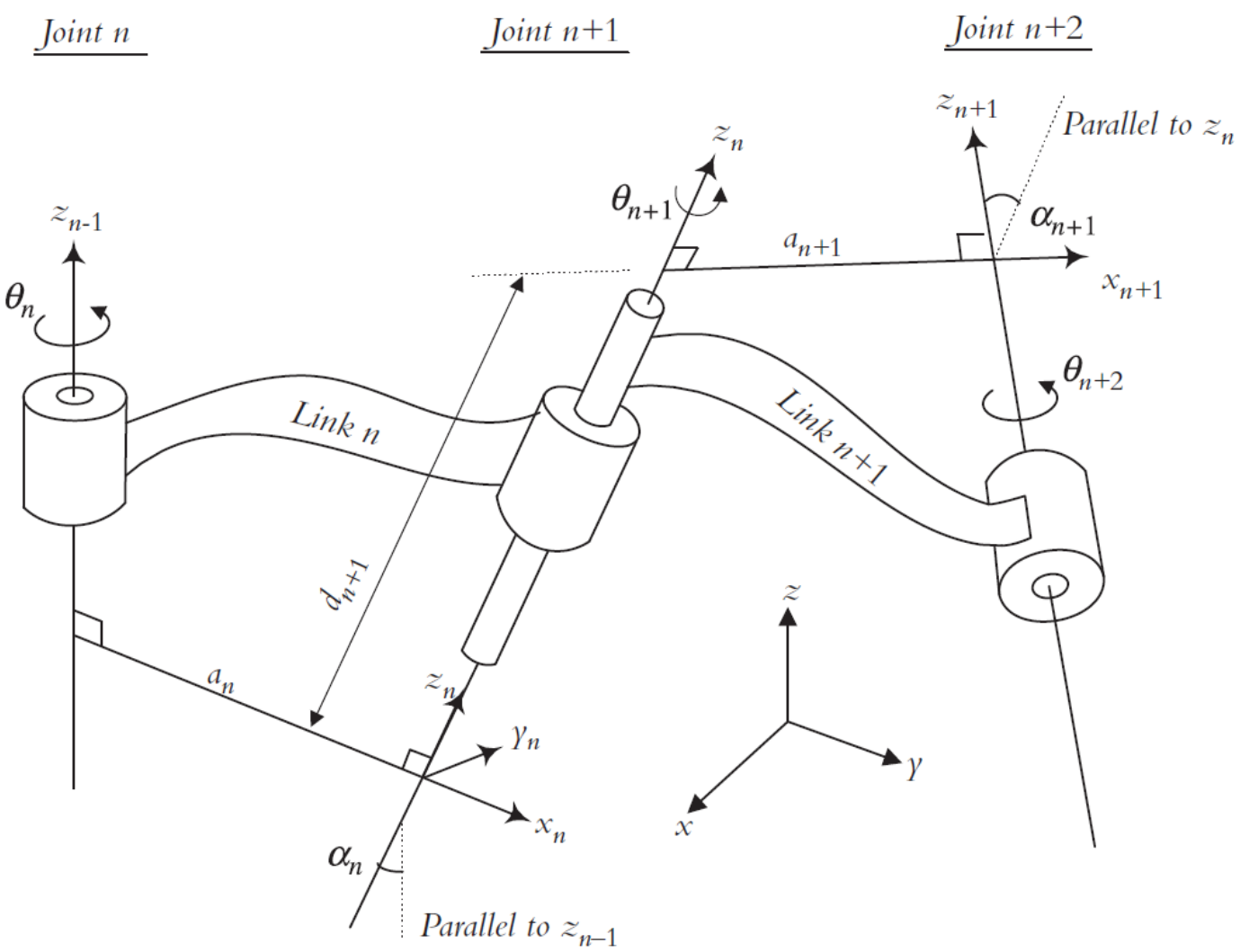


DH parameters

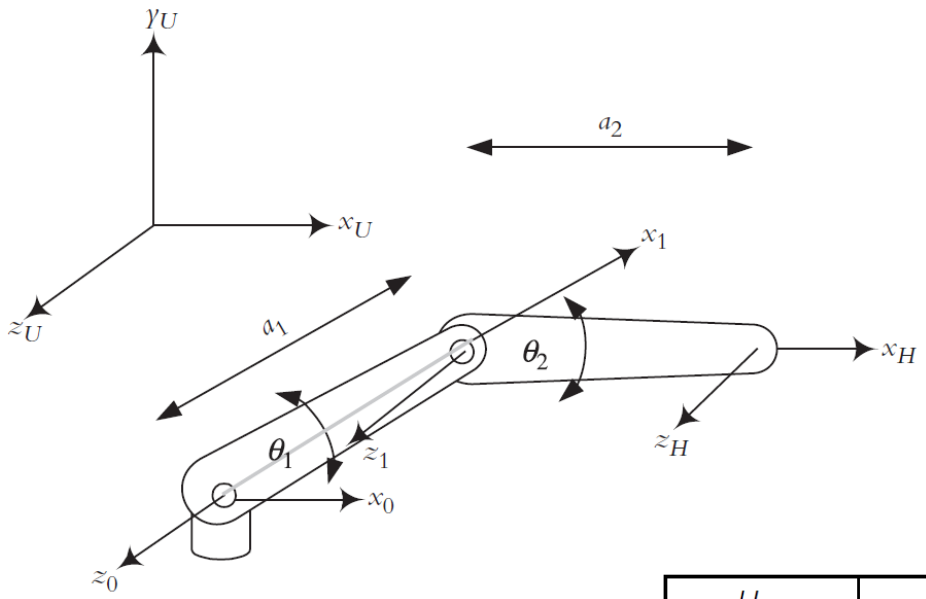
Parameter	Symbol	Meaning
Link Length	a_i	Distance from Z_{i-1} to Z_i along X_i
Link Twist	α_i	Angle between Z_{i-1} and Z_i about X_i
Link Offset	d_i	Distance from X_{i-1} to X_i along Z_i
Joint Angle	θ_i	Angle between X_{i-1} and X_i about Z_i



General arrangement of links and joints

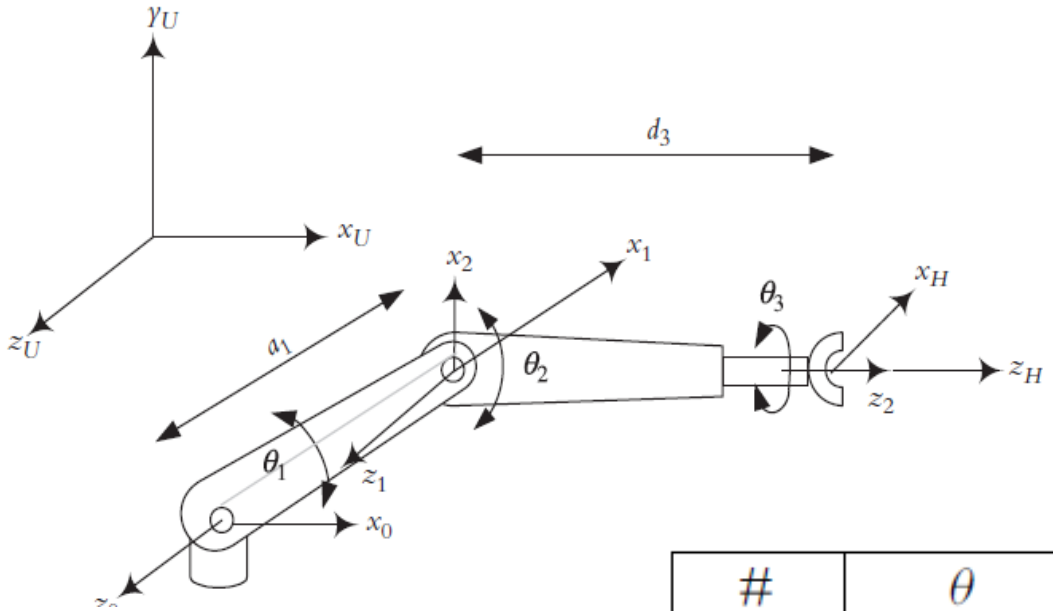

Joint n
Joint n+1
Joint n+2
z_{n-1}
θ_n
Link n
d_{n+1}
a_n
z_n
γ_n
x_n
α_n
Parallel to z_{n-1}
z_n
θ_{n+1}
a_{n+1}
z_{n+1}
Parallel to z_n
α_{n+1}
x_{n+1}
θ_{n+2}
Link n+1
z
y
x

Assign the necessary frames to the robot of Figure and derive the forward kinematic equation of the robot.



#	θ	d	a	α
0–1	θ_1	0	a_1	0
1–H	θ_2	0	a_2	0

Assign the necessary frames to the robot of Figure and derive the forward kinematic equation of the robot.



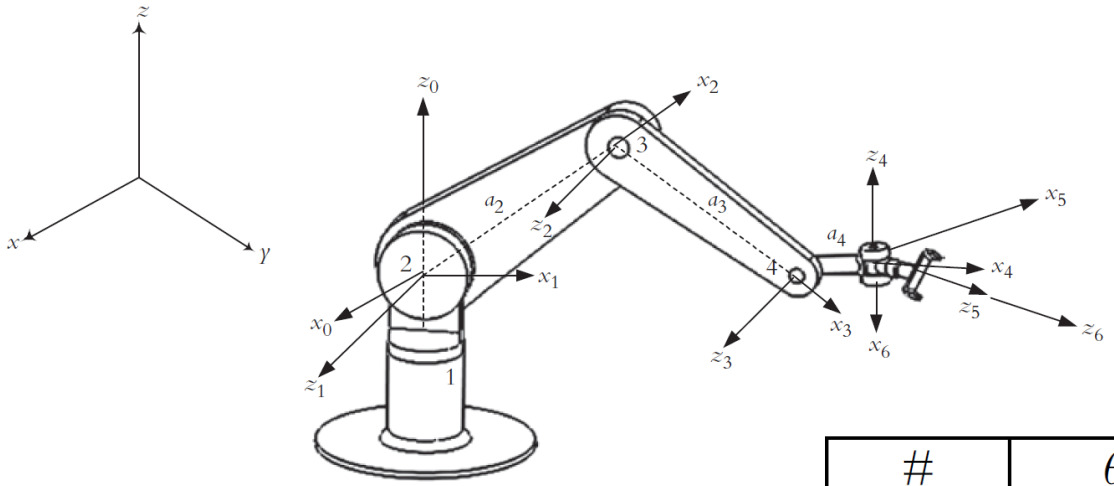
$$Rot(z,\theta) = \begin{bmatrix} cos\,\theta & -sin\,\theta & 0 \\ sin\,\theta & cos\,\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

#	θ	d	a	α
0–1	θ_1	0	a_1	0
1–2	$90 + \theta_2$	0	0	90
2–H	θ_3	d_3	0	0

$$A_1 = \begin{bmatrix} C_1 & -S_1 & 0 & a_1 C_1 \\ S_1 & C_1 & 0 & a_1 S_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_2 = \begin{bmatrix} -S_2 & 0 & C_2 & 0 \\ C_2 & 0 & S_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3 = \begin{bmatrix} C_3 & -S_3 & 0 & 0 \\ S_3 & C_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

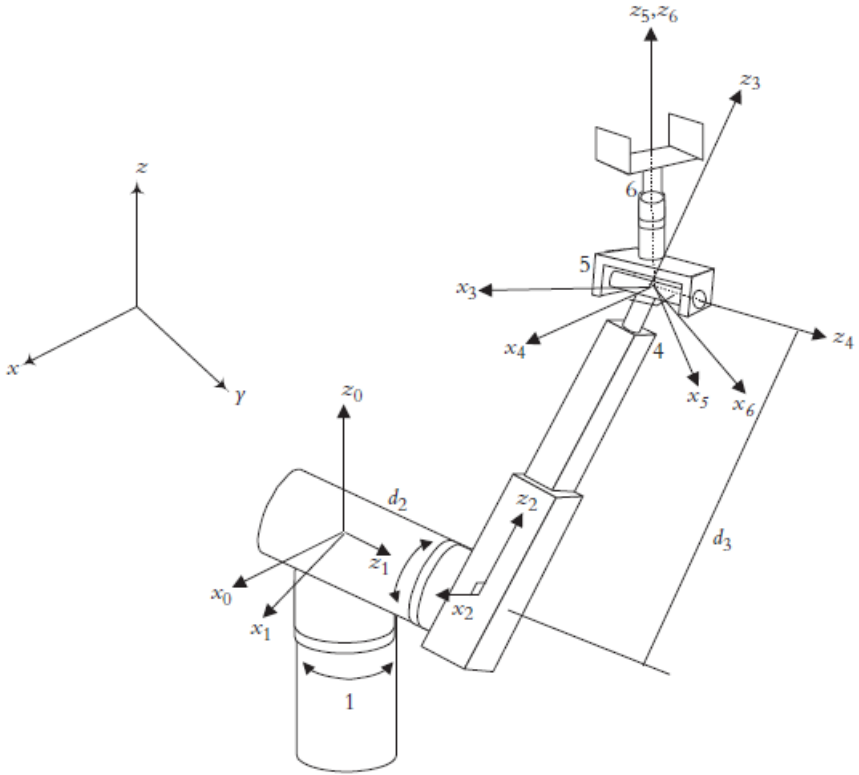
$$^0T_H = A_1 A_2 A_3$$

Assign the necessary frames to the robot of Figure and derive the forward kinematic equation of the robot.



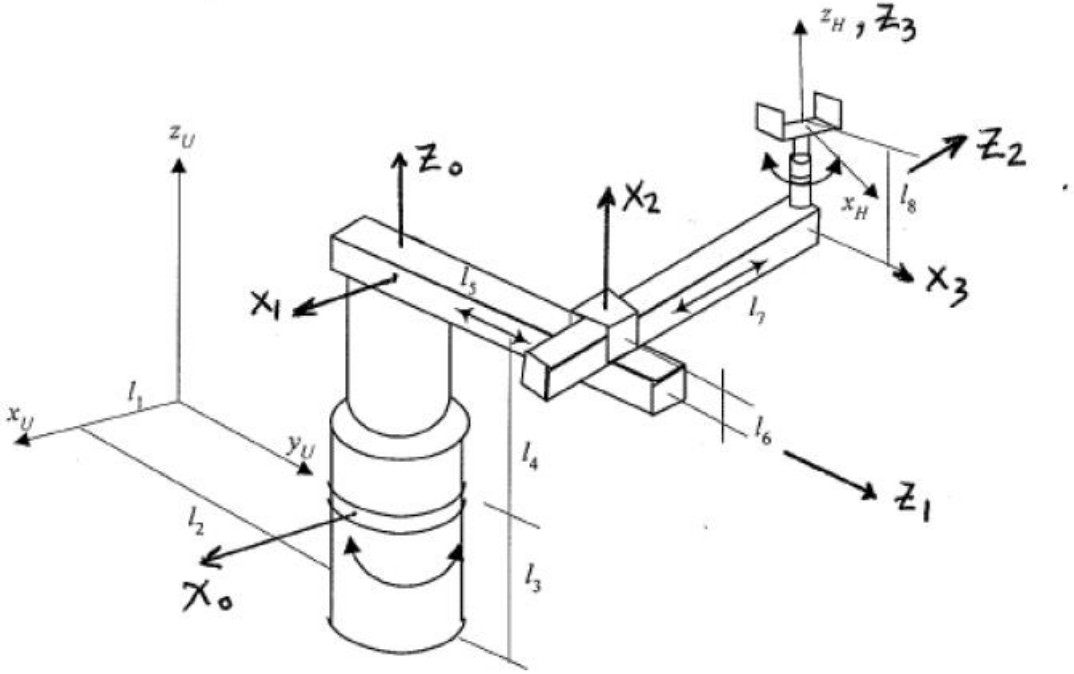
#	θ	d	a	α
0–1	θ_1	0	0	90
1–2	θ_2	0	a_2	0
2–3	θ_3	0	a_3	0
3–4	θ_4	0	a_4	−90
4–5	θ_5	0	0	90
5–6	θ_6	0	0	0

Assign the necessary frames to the robot of Figure and derive the forward kinematic equation of the robot.



#	θ	d	a	α
0–1	θ_1	0	0	−90
1–2	θ_2	d_2	0	90
2–3	0	d_3	0	0
3–4	θ_4	0	0	−90
4–5	θ_5	0	0	90
5–6	θ_6	0	0	0

Assign the necessary frames to the robot of Figure and derive the forward kinematic equation of the robot.



#	θ	d	a	α
0-1	θ_1	0	0	90
1-2	0	l_5	0	90
2-3	0	l_7	0	90
3-H	θ_4	0	0	0

Formulation transformation matrix from DH table

- **θ_i (Joint Angle):** Rotation about the previous $Z_{\{i-1\}}$ axis.
- **d_i (Joint Offset):** Translation along the previous $Z_{\{i-1\}}$ axis.
- **a_i or r_i (Link Length):** Translation along the X_i axis.
- **α_i (Link Twist):** Rotation about the X_i axis.

$$T_{i-1}^{i} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i)\cos(\alpha_i) & \sin(\theta_i)\sin(\alpha_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i)\cos(\alpha_i) & -\cos(\theta_i)\sin(\alpha_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Time for Discussions





Thank You!